



IMPLEMENTACIÓN TRANSFORMER

73.69 - LARGE LANGUAGE MODELS - 2026

Ing. Eugenia Piñeiro, Ing. Marina Fuster

TABLA DE CONTENIDOS

01. LANGCHAIN

—

02. DEMOS

—

03. FRAMEWORKS Y
APIs

—

04. CLOUD
COMPUTING

—

05. BIBLIOGRAFÍA

—

REPASO

- ¿Cuál es el objetivo principal de RAG?
- ¿Cuáles son sus principales componentes? ¿En qué consiste el indexing pipeline?
- ¿Qué bases de datos vectoriales vimos?
- ¿Cuál es el objetivo de Advanced RAG? ¿Qué técnicas recordás?

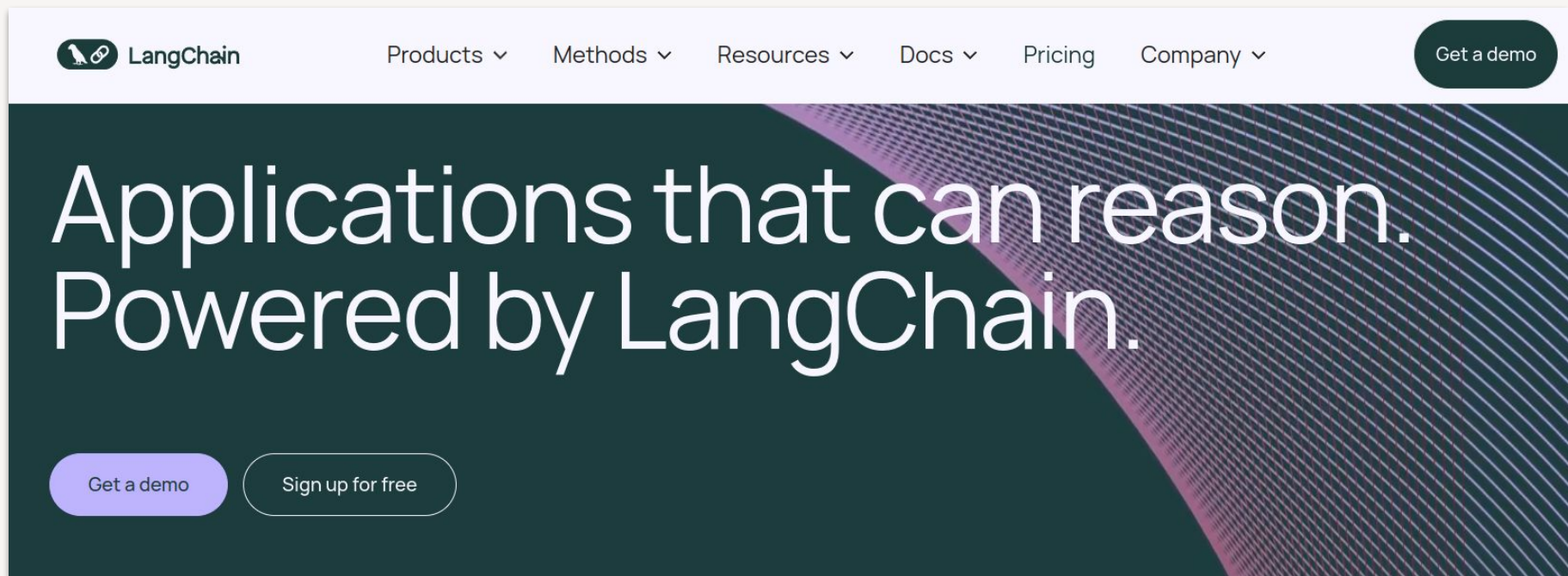
01
—

LANGCHAIN

Orchestration Framework

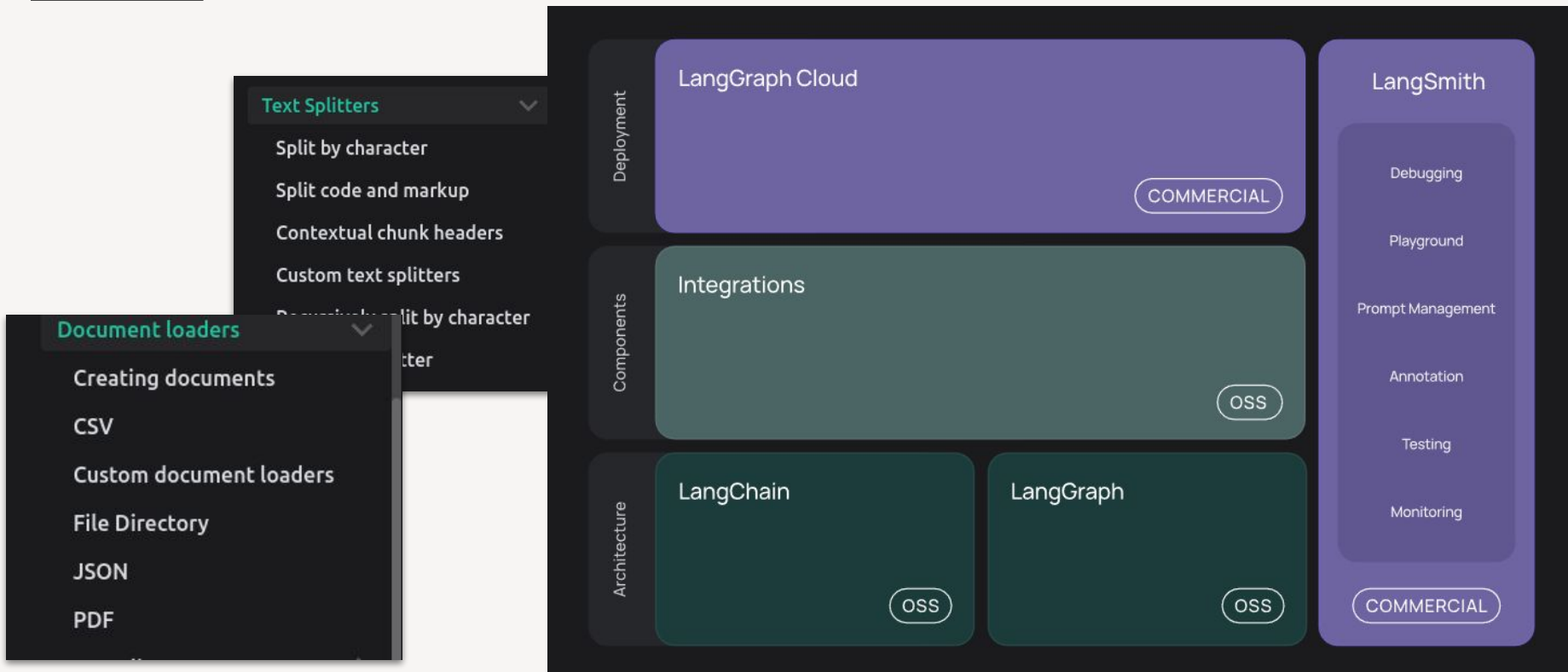
LANGCHAIN

Orchestration Framework para el desarrollo de aplicaciones con LLMs

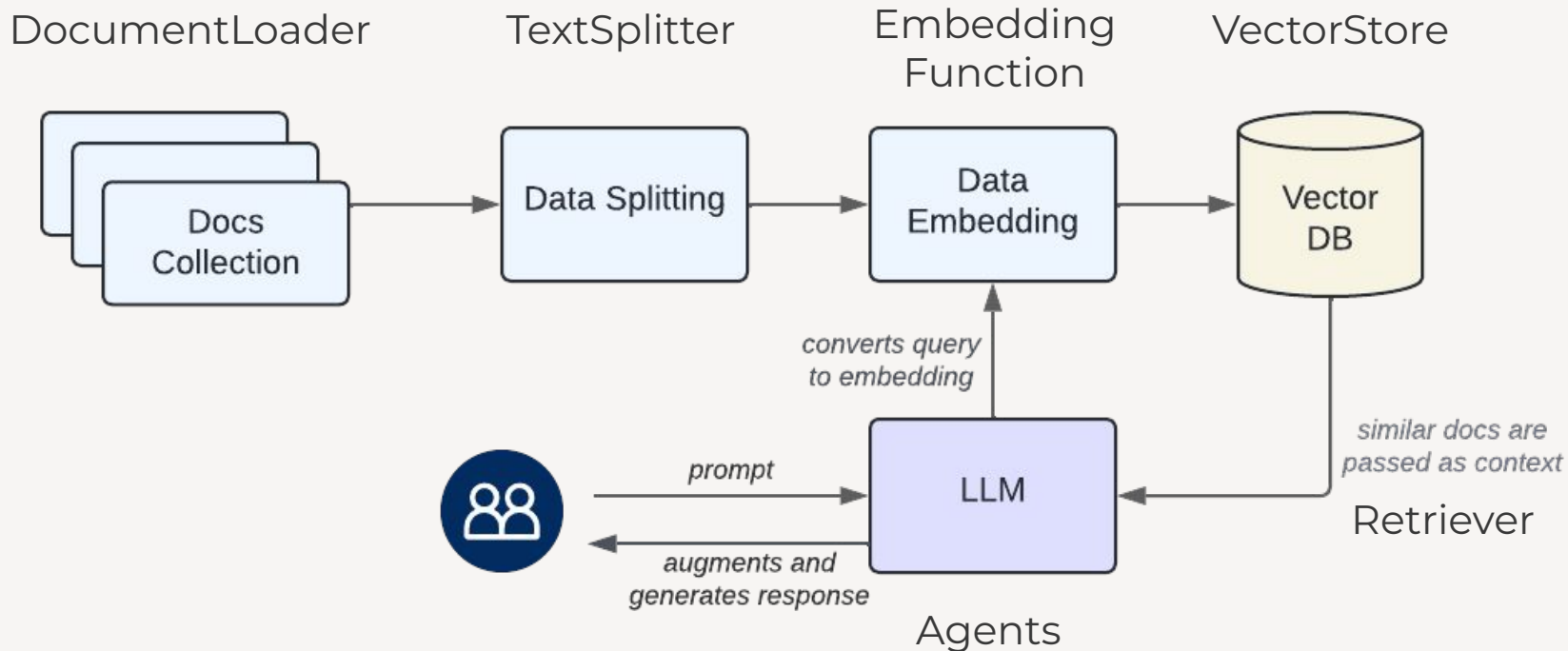


LANGCHAIN

[Langchain Doc](#)



LANGCHAIN

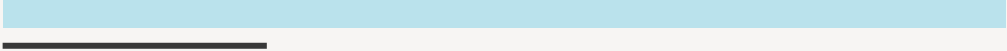


02

DEMOS

Splitter, BD, RAG, Advanced RAG

¿Cómo implemento RAG con Langchain ?

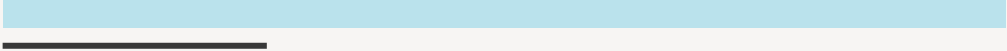
A decorative graphic consisting of two horizontal lines. The top line is a solid light blue bar. The bottom line is a thin dark blue line that starts at the left edge of the blue bar and extends to the right, ending slightly before the right edge of the blue bar.

02

DEMOS

Splitter, BD, RAG, Advanced RAG

¿Cómo implemento RAG con Langchain ?

A decorative graphic consisting of two horizontal lines. The top line is a solid light blue bar. The bottom line is a thin dark blue line that starts at the left edge of the blue bar and extends to the right, ending slightly before the blue bar's right edge.

DEMOS



[Langchain - Pinecone.ipynb](#)

[Langchain - ChromaDB.ipynb](#)

¿Cómo genero mi dataset,
lo divido y lo persisto en
una Bases de Datos?

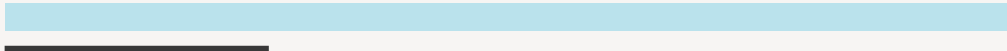
DEMOS



[Langchain - Virtual Assistant.ipynb](#)

¿Cómo genero una
respuesta utilizando el
LLM junto con RAG?

¿Cómo podemos visualizar
los embeddings?

A decorative graphic consisting of a thick light blue horizontal bar and a thin dark blue horizontal bar, both spanning the width of the text above them.

UMAP

Uniform Manifold Approximation and Projection

- Preserva relaciones no lineales
- Preserva clusters
- Identificar outliers/ruido
- Escalable a grandes datasets

Se construye un **grafo** del dataset

- Nodes → data point
- Edges → conectados según similitud

UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction

Leland McInnes

Tutte Institute for Mathematics and Computing
leland.mcinnnes@gmail.com

John Healy

Tutte Institute for Mathematics and Computing
jchealy@gmail.com

James Melville

jlmelville@gmail.com

September 21, 2020

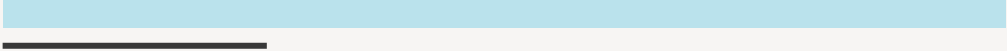
Abstract

UMAP (Uniform Manifold Approximation and Projection) is a novel manifold learning technique for dimension reduction. UMAP is constructed from a theoretical framework based in Riemannian geometry and algebraic topology. The result is a practical scalable algorithm that is applicable to real world data. The UMAP algorithm is competitive with t-SNE for visualization quality, and arguably preserves more of the global structure with superior run time performance. Furthermore, UMAP has no computational restrictions on embedding dimension, making it viable as a general purpose dimension reduction technique for machine learning.

1 Introduction

Dimension reduction plays an important role in data science, being a fundamental technique in both visualisation and as pre-processing for machine

¿Cómo puedo mejorar
la respuesta?



GUARDRAILS

Conjunto de **restricciones** de seguridad para monitorear y limitar la interacción user-LLM.

Prevenir

- Lenguaje ofensivo
- Desinformación
- *Contenido fuera de contexto*
- Problemas éticos
- Problemas legales
- ... entre otros.

¿CÓMO?

Técnicas

- *Prompt Engineering*
- RLHF
- Compliance check
- Predefined responses
- ... entre otras.

DEMOS



[Langchain - Advanced RAG.ipynb](#)

¿Cómo puedo mejorar los
resultados utilizando
técnicas de
Advanced RAG?

¿Qué más se puede investigar?

- Langchain
 - Models ChatCompletions
 - ConversationBufferMemory
 - Prompt Templates
 - Output Parser
 - Chains
- Poetry
- Postgres + pgvector extension

EXTRA



[Langchain - Text Splitter.ipynb](#)

¿Cómo funciona el
splitting?

03

OTROS FRAMEWORKS Y APIs

Pytorch y Tensorflow

HUGGING-FACE

Existen distintos modelos de Transformers: GPT-2/3/4, BERT, RoBERTa, DistilBERT, T5

Transformers

Q Search documentation Ctrl+K

V4.35.0 EN 114,893

GET STARTED

Transformers

Quick tour

Installation

TUTORIALS

Run inference with pipelines

Write portable code with AutoClass

Preprocess data

Fine-tune a pretrained model

 **Transformers**

State-of-the-art Machine Learning for [PyTorch](#), [TensorFlow](#), and [JAX](#).

 Transformers provides APIs and tools to easily download and train state-of-the-art pretrained models. Using pretrained models can reduce your compute costs, carbon footprint, and save you the time and resources required to train a model from scratch. These models support common tasks in different modalities, such as:

 **Natural Language Processing:** text classification, named entity recognition, question answering, language modeling, summarization, translation, multiple choice, and text generation.

 **Computer Vision:** image classification, object detection, and segmentation.

 **Audio:** automatic speech recognition and audio classification.

 **Multimodal:** table question answering, optical character recognition, information extraction from scanned documents, video classification, and visual question answering.

DEEP LEARNING FRAMEWORKS

Build, train y deploy de modelos de Deep Learning



04

CLOUD COMPUTING

¿Cómo puedo deployar mi modelo?

MLOps CLOUD

PRODUCT	aws	Microsoft Azure	Google Cloud Platform
Machine learning	Amazon SageMaker	Azure Machine Learning	Google Cloud AI Platform
Image recognition	Amazon Rekognition	Azure Cognitive Services	Google Cloud Vision
Speech	Amazon Polly, Amazon Transcribe	Azure Cognitive Services	Google Cloud Speech-to-Text and Text-to-Speech
Natural language processing	Amazon Comprehend	Azure Cognitive Services	Google Cloud Natural Language API:
Big Data Analytics	Databricks	Databricks	Databricks
Chat Bot	AWS Chatbot	Azure Bot Service	Dialogflow

☰ Azure



Sign in

Hugging Face on Azure

Deploy tens of thousands of pretrained Hugging Face Transformers models in Azure Machine Learning using Hugging Face endpoints.

Try Azure for free

Already using Azure? Try Hugging Face on Azure >



Products Solutions Pricing Documentation Learn Partner Network AWS Marketplace Customer Enablement Events Explore More

Tools to build and scale generative AI applications

Innovate faster with new capabilities, a choice of industry-leading FMs, and infrastructure that pushes the envelope to deliver the highest performance while lowering costs.

[Explore more generative AI tools](#)

Service

Amazon Q

Customize this generative AI-powered assistant for the needs of your business

Service

Amazon Bedrock

Easily build and scale applications with LLMs, FMs, and generative AI tools

Service

Amazon SageMaker

Build, train, and deploy your own FM models at scale

05

BIBLIOGRAFÍA

BIBLIOGRAFÍA

Van der Maaten, L., & Hinton, G. (2008). **Visualizing data using t-SNE**. Journal of Machine Learning Research, 9, 2579-2605. Retrieved from <https://jmlr.org/papers/volume9/vandermaaten08a/vandermaaten08a.pdf>

McInnes, L., Healy, J., & Melville, J. (2020). **UMAP: Uniform Manifold Approximation and Projection for dimension reduction**. Tufts Institute for Mathematics and Computing. Retrieved from <https://arxiv.org/pdf/1802.03426>

Aeonaten. (2020, September 21). **Understanding UMAP: Uniform Manifold Approximation and Projection**. Medium <https://medium.com/@aeonaten/understanding-umap-uniform-manifold-approximation-and-projection-cede51c477d9>